MIL-C-47118A(MI)
26 September 1979
Superseding
MIL-C-47118(MI)
24 May 1974

MILITARY SPECIFICATION

COATING, CONDUCTIVE LACQUER

This specification is approved for use by the US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a conductive lacquer surface coating for RFI attenuation capabilities.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of the invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

MIL-STD-1188	Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and publications required by contractors, in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: DRSMI-ESD(R&D), Redstone Arsenal, AL 35809, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8010

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing Materials

ASTM D-257 Test for D-C resistance or conductance of insulating materials.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. When specified by the procuring activity (see 6.2), a preproduction sample shall be prepared using the same methods proposed for subsequent production and shall be subjected to preproduction inspection (see 4.3).

3.2 Material. The material shall be a nitrocellulose lacquer with a silver filler for conductivity.

3.3 Property requirements.

3.3.1 Surface resistivity. The surface resistivity shall be .05 ohms/square, maximum.

3.3.2 Density. The density shall be 1.52 to 1.60 grams per cubic centimeter at 75 plus or minus 2 degrees Fahrenheit.

3.3.3 Solids content. The solids content shall be 52.5 to 56.5 percent by weight.

3.3.4 Settling. There shall be no more settling than will permit the lacquer to be stirred readily to a smooth, uniform mixture.

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3.3.5 Workmanship. The workmanship shall be such as to insure a product which is uniform and in conformance with this specification. The lacquer shall be free of dirt, foreign material or other contaminates.

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification and tests. The testing of the material shall be classified as follows:

4.2.1 Preproduction inspection (4.3)

4.2.2 Quality conformance inspection. (4.4)

4.3 Preproduction inspection. Preproduction inspection shall consist of all examinations and tests specified herein and shall be conducted only on preproduction samples. The preproduction samples shall be prepared using the same methods to be used for the manufacturer of all subsequent production material.

4.4 Quality conformance inspection.

4.4.1 Quality conformance inspection tests shall be the examination and tests required for each production lot. A lot shall be defined as that quantity of material which has been manufactured at one time by a chemical or physical process.

4.4.2 Sampling. A random sample of sufficient quantity to complete the required testing shall be obtained from each production lot.

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4.4.3 Quality conformance tests shall include the tests and requirements specified per 3.3.2, 3.3.3, and 3.3.5. Failure of any of the tests shall be cause for rejection of the production lot.

4.5 Test methods.

4.5.1 Examination of product. The coating shall be examined for conformance with the material and workmanship requirements of this specification.

4.5.2 Surface resistivity. To determine conformance to 3.3.1, the surface resistivity shall be determined by use of a volt-ohm meter using the X1 ohm scale.

4.5.2.1 Test specimen preparation. A minimum of three determinations per test shall be made for qualification. The test specimen shall consist of a clean glass slide coated with the conductive lacquer.

4.5.2 Surface resistivity. To determine conformance to 3.3.1, the surface resistivity shall be determined in accordance with ASTM D-257. The resistance shall be determined by use of a volt-ohm meter using the X1 ohm scale.

4.5.2.2 Coating. Coat one side of the glass with the silver filled lacquer to a thickness of .001 inch minimum. Coating may be applied by brushing or spraying.

4.5.2.3 Drying. Air dry the coated slide at room temperature for one hour minimum.

4.5.2.4 Completing cure. Complete the cure by placing the slide in an oven at 150 degrees plus or minus 5 degrees Fahrenheit for one hour.

NOTE: Butyl acetate should be used as the thinner.

4.5.3 Density. To determine conformance to 3.3.2, the density shall be determined at 75 plus or minus 2 degrees Fahrenheit, using a 12 cubic centimeter pycnometer.

4.5.4 Solids content. To determine conformance to 3.3.3, the solids content shall be determined as follows: weigh 3.0 plus or minus .5 grams (to the nearest .001 gram) into a tared flat bottom dish approximately 8 centimeter in diameter. Heat to a constant weight in an oven at 105 plus or minus 5 degrees Fahrenheit. Place in a desiccator and reweigh to the nearest .001 gram.

Calculate percent solids as S/W x 100

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Where S = weight of sample after heating

Where W = weight of sample

4.5.5 Settling. To determine conformance to 3.3.4, settling shall be determined as follows: Stir the contents of the container and examine the paint for uniformity of dispersion. Settling is defined as the material which has settled to the bottom and cannot readily be stirred into the mixture.

5. PACKAGING

5.1 Preservation, packing, unitization, and marking - Commercial packaging (see 6.2).

5.1.2 Commercial. Preservation, packing, unitization, and marking shall be in accordance with MIL-STD-1188.

5.1.3 In addition the following markings are required:

a. Date of manufacture (month and year).

b. Shelf life expiration date.

6. NOTES

6.1 Intended use. This material is to be used as a conductor for RF purposes.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether preproduction samples and test reports are required.

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c. Special packaging and packing instructions, if required.

d. Other special instructions.

e. Selection of applicable level of preservation, packaging, and packing.

6.3 Supersession data. This specification includes the requirements of Missile Iterim Specification MIS-15741, dated 16 May 1966.

Custodian:
Army-MI

Preparing activity:
Army-MI

Project No. 8010-A154

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-C-47118A(MI) COATING, CONDUCTIVE LACQUER

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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